

Azure Control Plane Monitoring & Insider Threat Detection

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Executive Summary

This project demonstrates hands-on experience monitoring Azure control plane activity and analyzing insider risk scenarios from both a Security Operations Center (SOC) and Governance, Risk, and Compliance (GRC) perspective. Using a free-tier Azure environment, the project focuses on identity lifecycle events, role-based access control (RBAC) changes, and audit visibility—key areas frequently abused in real-world cloud security incidents.

Rather than weakening built-in protections, this project intentionally preserves strong security controls (such as mandatory MFA) and documents how those controls deter low-effort or low-value attacker behavior. Findings are analyzed through a SOC investigation lens and mapped to industry frameworks including NIST SP 800-53 and ISO/IEC 27001.

Project Objectives

- Validate default Azure control plane logging
- Simulate realistic insider risk behaviors without bypassing security controls
- Analyze administrative activity as a SOC analyst
- Map technical findings to GRC frameworks employers recognize

Environment Overview

- Platform: Microsoft Azure (Free Tier)
- Identity Provider: Microsoft Entra ID
- Logging Source: Azure Activity Log
- Security Posture: Security Defaults enabled (mandatory MFA)

Baseline Logging Validation

Objective

Establish visibility into Azure's default control plane logging and understand what administrative actions are recorded in a new subscription.

Actions Performed

- Reviewed Azure Monitor → Activity Log
- Confirmed default retention and event categories
- Identified absence of logs prior to administrative activity

Key Observations

Azure Activity Logs are empty until control plane actions occur. Therefore, no synthetic or baseline administrative events are generated.

Activity Log showing no significant events prior to testing:

The screenshot displays the Azure Monitor 'Activity log' interface. The left sidebar shows navigation options: Overview, Activity log (selected), Alerts, Issues (preview), Metrics, Logs, Change Analysis, and Entra health. The main panel has a top bar with a search box and various action buttons (Activity, Edit columns, Refresh, Export Activity Logs, Download as CSV, Insights, Feedback, Pin current filters, Reset filters). Below this is a banner for Log Analytics. The main content area shows filters: Management Group: None, Subscription: Azure subscription 1, Event severity: All, Timespan: Last 6 hours, and an Add Filter button. It indicates '1 item.' and displays a table with columns: Operation name, Status, Time, Time stamp, Subscription, and Event initiated by. The table contains one entry: 'Create subscription' with status 'Succeeded', time '34 minutes ...', time stamp 'Fri Jan 23 2...', subscription 'Azure subscription 1', and a redacted 'Event initiated by' field.

Operation name	Status	Time	Time stamp	Subscription	Event initiated by
Create subscription	Succeeded	34 minutes ...	Fri Jan 23 2...	Azure subscription 1	[Redacted]

Simulated Insider Behavior

Objective

Generate realistic, high-signal administrative events commonly associated with insider misuse.

Actions Performed

1. Created a new Entra ID user (`temp.insider`)
2. Assigned elevated RBAC roles at the subscription level
3. Removed roles shortly after assignment
4. Preserved mandatory MFA enforcement and did not bypass security controls

Rationale

These actions simulate:

- Unauthorized account provisioning
- Privilege escalation
- Short-lived elevated access (a common insider tactic)

New user created:

Create new user ...

Create a new internal user in your organization

BasicsPropertiesAssignmentsReview + create

Basics

User principal name

temp.insider[redacted]

Display name

Temp Insider

Mail nickname

temp.insider

Password

[redacted]

Account enabled

Yes

Properties

User type

Member

Assignments

Administrative units

Groups

Roles

Create

< Previous

Next >

Give feedback

Role assignment event:

Add role assignment ...

Role Members Conditions Review + assign

Role AgFood Platform Service Contributor

Scope [Redacted]

Members

Name	Object ID	Type
Temp Insider	[Redacted]	User

Description No description

Role removal event:

Azure subscription 1 | Access control (IAM) ☆ ...

Subscription

Search [] ◊ << + Add ▾ ⬇ Download role assignments ⚙ Edit columns ↻ Refresh 🗑 Delete 🗨 Feedback

Overview
Activity log
Access control (IAM)
Tags
Diagnose and solve problems
Security
Resource visualizer
Events
Resource groups
Resources
Cost Management
Billing
Settings
Monitoring
Help

Remove role assignments

Are you sure you want to remove the selected role assignment?

[View assignments](#)

Search by name or email [] Type: All Role: All Scope: All scopes Group by: Role

All (3) Job function roles (1) Privileged administrator roles (2)

☒	Name ↑↓	Type ↑↓	Role ↑↓	Scope ↑↓	Condition ↑↓
▽	Owner (2)				
☐	[Redacted]		Owner	This resource	Add
☐	[Redacted]		Owner	This resource	Add
▽	AgFood Platform Service Contributor (1)				
☒	Temp Insider 462c9bdd-e941-41e7-8de0-9... User		AgFood Platform Service Contributor	This resource	None

Step 3 — SOC Analyst Investigation

Objective

Analyze generated events as a SOC analyst would during an investigation.

Investigation Questions

- Who initiated the action?
- What role or privilege level did they have?
- Was the behavior expected or justified?
- What indicators would raise concern?

Key Findings

- All privileged actions were attributable via the "Initiated by" identity field
- Rapid role assignment and removal represents elevated insider risk
- Mandatory MFA prevented further unauthorized activity by the test user

SOC Analysis Table

Event	Risk	Detection Source	Recommended Control
User creation	Unauthorized access	Entra ID Audit Logs	Approval workflows
Role assignment	Privilege escalation	Azure Activity Log	RBAC least privilege
Rapid role removal	Covering tracks	Azure Activity Log	Time-based alerts
MFA enforcement	Attack deterrence	Security Defaults	Maintain MFA

Activity Log event details:

Monitor | Activity log

Microsoft

Search

Activity

Edit columns

Refresh

Export Activity Logs

Download as CSV

Insights

Feedback

Pin current filters

Reset filters

Overview

Activity log

Alerts

Issues (preview)

Metrics

Logs

Change Analysis

Service health

Workbooks

Dashboards with Grafana

Looking for Log Analytics? In Log Analytics you can search for performance, diagnostics, health logs, and more. [Visit Log Analytics](#)

Search

Quick Insights

Management Group : None

Subscription : Azure subscription 1

Event severity : All

Timespan : Last 6 hours

Add Filter

3 items.

Operation name	Status	Time	Time stamp	Subscription	Event initiated by
> Delete role assignment	Succeeded	2 minutes a...	Fri Jan 23 2...	Azure subscription 1	
> Create role assignment	Succeeded	10 minutes ...	Fri Jan 23 2...	Azure subscription 1	
Create subscription	Succeeded	an hour ago	Fri Jan 23 2...	Azure subscription 1	

JSON Log:

Create role assignment

Fri Jan 23 2026 18:10:00 GMT-0500 (Eastern Standard Time)



+ New alert rule

Summary

JSON

Change history (Change history is not applicable to this type of activity log)

```
1 {
2   "authorization": {
3     "action": "Microsoft.Authorization/roleAssignments/write",
4     "scope": "/subscriptions/8af39c14-3ec2-4845-9ebc-5abcc0dc0964/providers/Microsoft.
    Authorization/roleAssignments/e6795548-4fab-46a9-8144-38bdc891b801"
5   },
6   "caller": "XXXXXXXXXX@hotmail.com",
7   "channels": "Operation",
8   "claims": {
9     "aud": "https://management.core.windows.net/",
10    "iss": "https://sts.windows.net/b16f6043-95b7-4940-8a3f-8b6de876c83b/",
11    "iat": "1769207404",
12    "nbf": "1769207404",
13    "exp": "1769212532",
14    "http://schemas.microsoft.com/claims/authnclassreference": "1",
15    "acrs": "p1",
```

GRC Control Mapping

Objective

Translate technical findings into governance and compliance language used by security leadership.

NIST SP 800-53 — AC-2 (Account Management)

Observation: User accounts and role changes were fully auditable. Elevated access was granted and removed without approval controls.

Assessment: Control partially implemented.

Recommendation: Implement role assignment approvals and periodic access reviews.

NIST SP 800-53 — AU-2 (Audit Events)

Observation: Azure Activity Logs captured privilege changes with sufficient attribution. Some tenant-level security posture changes lacked consistent audit visibility.

Assessment: Control partially implemented.

Recommendation: Centralize logs and define required audit events.

ISO/IEC 27001 — Annex A.9 (Access Control)

Observation: RBAC and MFA effectively limited insider misuse. Privileged access governance was informal.

Assessment: Control implemented with gaps.

Recommendation: Formalize privileged access management policies.

Security Control Effectiveness Note

Mandatory MFA enforcement significantly increased attacker friction. For a low-value account, the additional effort required to enroll authentication factors acted as an effective deterrent. This reflects real-world attacker behavior, where high-friction controls often cause attackers to abandon targets rather than escalate effort.

Conclusion

This project demonstrates practical cloud security skills across technical detection, SOC investigation, and GRC mapping. It highlights the importance of preserving strong security controls during testing and shows how meaningful security insights can be derived without bypassing protections.

The approach and findings closely mirror real-world SOC and cloud security operations, making this project directly applicable to IT, SOC, and security analyst roles.

Key Skills Demonstrated

- Azure control plane monitoring
- Identity and access management analysis
- Insider threat detection concepts
- Audit log interpretation
- NIST and ISO control mapping
- Security-first decision making